

# Gauss–Markov Processes

## Filtering and Stochastic Differential Equations

As presented above, Gaussian process regression is a rather general formulation of inference on a function from finitely many data. However, computing the posterior's parameters associated with  $N$  real-valued data, in Eqs. (4.6) and (4.7), involves the inversion of a  $N \times N$  covariance matrix; hence the computational cost of doing so in general rises faster than  $N$  (general matrix inversion has cost  $\mathcal{O}(N^3)$ , see also §III). This nonlinear cost is an obstacle to the application of this framework for the design of efficient numerical algorithms. Ideally, a numerical routine should be able to keep running virtually indefinitely and increase its precision over time. This requires an inference framework whose computational complexity scales at most linearly,  $\mathcal{O}(N)$ . Parametric regression, introduced in §4.1, has this property; but it also has a finite model capacity, thus can only compute approximations of limited quality.

Some numerical problems have a trait that offers another way to address this challenge. They construct an ordered sequence of estimates by moving along a one-dimensional path parametrised by a variable  $t$  that, for intuition, we may as well call *time*. The central examples are:

**Quadrature:** Univariate integration methods estimate an integral  $F = \int_t f(t)dt$ , by evaluating  $f$  for various values  $t_i \in \mathbb{R}$ , stepping from one end of the integration domain to the other.

**ODEs:** Solvers for ordinary differential equations estimate a curve (a continuous function of univariate input)  $x(t), t \in \mathbb{R}$ , such that  $x'(t) = f(x(t), t)$ . In doing so they evaluate the multivariate function  $f$  for various values of  $t$  and  $x$ . (Often

the sequence of values  $t_i$  is ordered). Of course, such solvers can be thought of as solving a univariate integration problem, where the integration path is itself an (unknown) curve

**Nonlinear optimisation:** Algorithms that search for local extrema  $x_* = \arg \min f(x)$  of a real-valued differentiable function  $f : \mathbb{R}^N \rightarrow \mathbb{R}$  usually construct a sequence of estimates  $x_t \in \mathbb{R}^N$  designed to hopefully converge to the true  $x_*$ . As in the ODE setting, the estimates themselves are multivariate; but the sequence of estimates can be thought of as lying on a (discretised) curve. This notion is less precise than the two above, since it is not always clear how to account for varying step-lengths  $x_t - x_{t-1}$ . The locations  $x_t$  actually lie in a high-dimensional vector space with a natural Euclidean metric, so the “times” separating the points  $x_t$  are subject to some subtle scaling. Nevertheless, the notion of a time-series will turn out to be helpful anyway in §IV.

## ► 5.1 Markov Chains and Message Passing on Chain Graphs

Parametric regression models keep a finite global memory. But in the above settings, with observations on an ordered one-dimensional space, we may also try to construct models that pass a finite amount of information forward and backward along the line. This notion of a finite memory through time is formalised by the *Markov property*.<sup>1</sup>

<sup>1</sup> Markov (1906)

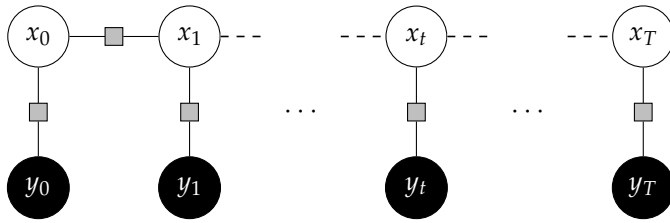
**Definition 5.1** (Markov chain). *Consider a discrete set of latent variables (called states)  $X = [x_t]_{t=0,\dots,T}$  with joint probability density  $p(X)$ . Then  $p$  is a Markov chain if it has the Markov property that  $x_t$  is conditionally independent of all preceding states given the direct precursor  $x_{t-1}$ :*

$$p(x_t \mid x_0, x_1, \dots, x_{t-1}) = p(x_t \mid x_{t-1}). \quad (5.1)$$

In this sense, the states collected in  $x_t$  can be thought of as a kind of sufficient local memory. If we know the state at any point in time, we know all that is possible to know about the future behaviour of the system at time  $t$ . In addition, assume observations  $y_t$  that only directly relate to the local state  $x_t$ ,

$$p(y_t \mid X) = p(y_t \mid x_t).$$

These two structural restrictions on a probabilistic model suffice to make inference on the latent variables linear in  $T$ . A first



**Figure 5.1:** Graphical model (factor graph) for a Markov chain. In this bipartite notation, every white node represents a latent variable, every black node an observable, and every grey box a factor in the joint probability distribution of  $(X, \mathbf{y})$ . Edges in the graph connect variables and observables to the factors in which they appear. The Markov property is visibly reflected in the fact that this graph is a *chain*. If all the factors are Gaussian, this is the generic form of a linear state space model. If they additionally all have the same parameters, it is a linear time-invariant model.

interesting observation is that it allows *recursive prediction*. Assume we have observed the *time series*  $\mathbf{y}_{0:t-1} := [y_0, \dots, y_{t-1}]$  and want to predict the states as described by  $p(x_t \mid \mathbf{y}_{0:t-1})$ . This posterior marginal distribution is given by

$$\begin{aligned} p(x_t \mid \mathbf{y}_{0:t-1}) &= \frac{\int_{j \neq t} p(X) p(\mathbf{y}_{0:t-1} \mid X) d\mathbf{x}_j}{\int p(X) p(\mathbf{y}_{0:t-1} \mid X) dX} \\ &= \frac{\int_{j \neq t} p(\mathbf{y}_{0:t-1} \mid X_{0:t-1}) p(x_0) d\mathbf{x}_0 \left( \prod_{0 < j < t} p(x_j \mid x_{j-1}) dx_j \right) p(x_t \mid x_{t-1}) \left( \prod_{i > t} p(x_i \mid x_{i-1}) dx_i \right)}{\int p(\mathbf{y}_{0:t-1} \mid X_{0:t-1}) p(x_0) d\mathbf{x}_0 \left( \prod_{0 < j < t} p(x_j \mid x_{j-1}) dx_j \right) p(x_t \mid x_{t-1}) dx_t \left( \prod_{i > t} p(x_i \mid x_{i-1}) dx_i \right)} \\ &= \frac{\int_{j < t} p(x_t \mid x_{t-1}) p(\mathbf{y}_{0:t-1} \mid X_{0:t-1}) p(x_0) d\mathbf{x}_0 \left( \prod_{0 < j < t} p(x_j \mid x_{j-1}) dx_j \right)}{\int_{j < t} p(\mathbf{y}_{0:t-1} \mid X_{0:t-1}) p(x_0) d\mathbf{x}_0 \left( \prod_{0 < j < t} p(x_j \mid x_{j-1}) dx_j \right)} \\ &= \int p(x_t \mid x_{t-1}) p(x_{t-1} \mid \mathbf{y}_{0:t-1}) dx_{t-1}. \end{aligned}$$

In other words, it is possible to iteratively compute a posterior over  $x_t$  given all previous observations, by an alternating sequence of *prediction-update* steps that only involve local quantities:

**Predict:** use the posterior from the previous step to compute<sup>2</sup>

$$p(x_t \mid \mathbf{y}_{0:t-1}) = \int p(x_t \mid x_{t-1}) p(x_{t-1} \mid \mathbf{y}_{0:t-1}) dx_{t-1}. \quad (5.2)$$

If  $t = 0$ , start the induction with the prior  $p(x_0)$ .

**Update:** include the local observation into the posterior by Bayes' theorem:

$$p(x_t \mid \mathbf{y}_{0:t}) = \frac{p(y_t \mid x_t) p(x_t \mid \mathbf{y}_{0:t-1})}{\int p(y_t \mid x_t) p(x_t \mid \mathbf{y}_{0:t-1}) dx_t}. \quad (5.3)$$

Expression (5.2) is known as the *Chapman–Kolmogorov* equation. It can be found as Eqs. (5) and (5\*) in a seminal paper by Kolmogorov (1936).

The situation is only slightly more complicated if, having collected all observations  $\mathbf{y} = \mathbf{y}_{0:T}$ , one subsequently has to compute a marginal distribution  $p(x_t \mid \mathbf{y})$  for some  $0 \leq t < T$ . Due

<sup>2</sup> These steps are a special case of a general algorithm called *belief propagation* or the *sum-product algorithm* (Pearl, 1988; Lauritzen and Spiegelhalter, 1988). It formalises the computational cost of inference in joint probability distributions given their factorisation into local terms. There are visual formal languages capturing such factorisation properties, known as *graphical models* (e.g. Figure 5.1).

to the Markov property,  $x_t$  is conditionally independent of later observations given  $x_{t+1}$ :

$$p(x_t | x_{t+1}, \mathbf{y}) = p(x_t | x_{t+1}, \mathbf{y}_{0:t}) \quad (5.4)$$

and of the form

$$\begin{aligned} &= \frac{p(x_t, x_{t+1} | \mathbf{y}_{0:t})}{p(x_{t+1} | \mathbf{y}_{0:t})} \\ &= \frac{p(x_{t+1} | x_t, \mathbf{y}_{0:t}) p(x_t | \mathbf{y}_{0:t})}{p(x_{t+1} | \mathbf{y}_{0:t})} \\ &= \frac{p(x_{t+1} | x_t) p(x_t | \mathbf{y}_{0:t})}{p(x_{t+1} | \mathbf{y}_{0:t})}. \end{aligned} \quad (5.5)$$

Using this, we can write the marginal posterior on  $x_t$  as follows<sup>3</sup>

$$\begin{aligned} p(x_t | \mathbf{y}) &= \int p(x_t, x_{t+1} | \mathbf{y}) dx_{t+1} \\ &= \int p(x_t | x_{t+1}, \mathbf{y}) p(x_{t+1} | \mathbf{y}) dx_{t+1} \\ &= \int p(x_t | x_{t+1}, \mathbf{y}_{0:t}) p(x_{t+1} | \mathbf{y}) dx_{t+1}. \end{aligned} \quad (5.6)$$

Then, we just insert Eq. (5.5) into Eq. (5.6),<sup>4</sup> which yields

$$p(x_t | \mathbf{y}) = p(x_t | \mathbf{y}_{0:t}) \int p(x_{t+1} | x_t) \frac{p(x_{t+1} | \mathbf{y})}{p(x_{t+1} | \mathbf{y}_{0:t})} dx_{t+1}. \quad (5.7)$$

So to compute the marginal  $p(x_t | \mathbf{y})$  for all  $0 \leq t \leq T$ , using the notion of time, we can think of this as first performing a *forward-pass* as described above to compute the predictions  $p(x_t | \mathbf{y}_{0:t-1})$  and updated beliefs  $p(x_t | \mathbf{y}_{0:t})$ . The final update step in that pass provides  $p(x_T | \mathbf{y})$ , from which we can start a *backward-pass* to compute the posterior marginals  $p(x_t | \mathbf{y})$  using Eq. (5.7) and all terms computed in the forward pass. This nomenclature, of *passing messages* along the Markov chain is popular in statistics and machine learning, and applies more generally to tree-structured graphs (i.e. not just chains).<sup>5</sup> In signal processing, where time series play a particularly central role, the forward pass is known as *filtering*, while the backward updates are known as *smoothing*. Figure 5.2 depicts the output of these methods. The next section spells out their computations in the case when the underlying state-space model is linear and Gaussian.

**Exercise 5.2 (easy).** Show that Eq. (5.4) holds. Hint: Use Bayes' theorem and the Markov property (5.1).

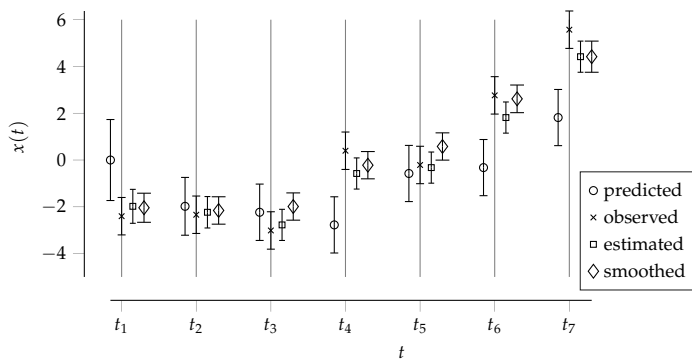
<sup>3</sup> This derivation is based on the exposition in §8.1 of Särkkä (2013)

<sup>4</sup> Kitagawa (1987)

<sup>5</sup> Bishop (2006), §8.4.2

## ► 5.2 Linear Gaussian State-Space Models

The abstract form of the message passing algorithm above becomes particularly elegant if all prior and conditional distributions in the model are Gaussian, involving only linear relations.



**Figure 5.2:** Linear time-invariant filtering and smoothing in a discrete time series. At each discrete time step (next to each other for legibility), the plot shows the predictive distribution (5.10), the observation with likelihood (5.9), the intermediate estimation (filtering) posterior (5.3), and the smoothed posterior (5.14). If the state-space model is linear and Gaussian, the estimation and smoothed posteriors are computable by the Kálmán filter and smoother, respectively. At each location, the variance drops from prediction to estimation belief, and from estimation to smoothed belief.

We will assume the following forms, using popular notational conventions for the linear maps involved:

$$\begin{aligned} p(x_0) &= \mathcal{N}(x_0; m_0, P_0), \\ p(x_{t+1} | x_t) &= \mathcal{N}(x_{t+1}; A_t x_t, Q_t), \\ p(y_t | x_t) &= \mathcal{N}(y_t; H_t x_t, R_t). \end{aligned} \quad (5.8) \quad (5.9)$$

The latter two relations are also often written, and known as, the *dynamic model* and an *measurement model*, respectively:

$$\begin{aligned} x_{t+1} &= A_t x_t + \xi_t \quad \text{with} \quad \xi_t \sim \mathcal{N}(0, Q_t) \quad (\text{dynamic}), \\ y_t &= H_t x_t + \zeta_t \quad \text{with} \quad \zeta_t \sim \mathcal{N}(0, R_t) \quad (\text{measurement}). \end{aligned}$$

In signal processing and control theory, the matrices  $A_t, H_t, Q_t, R_t$  are known as the *transition* and *measurement* matrices, *process noise* and *observation noise* covariances, respectively. We will sometimes use these intuitions evoking a dynamical system, but it will not always fit directly to the numerical domain. The entire setup is known as a *linear Gaussian system* in control theory. However, to avoid confusion with the regression models defined in §4.1 and §4.2 (which are also Gaussian and linear), we will use another popular convention and call this a *linear Gaussian state-space* model to stress that the inference takes place in terms of the time-varying states  $x_t$ . If the parameters are independent of time,  $A_t = A, Q_t = Q, H_t = H$  and  $R_t = R$ , for all  $t$ , they define a *linear time-invariant (LTI) system*.

Under these assumptions, the predictive, updated and posterior marginals constructed above all retain Gaussian form. The Chapman–Kolmogorov equation (5.2) becomes

$$\begin{aligned} p(x_{t+1} | y_{0:t}) &= \mathcal{N}(x_{t+1}; m_{t+1}^-, P_{t+1}^-), \quad \text{with} \quad (5.10) \\ m_{t+1}^- &= A_t m_t \quad \text{and} \quad P_{t+1}^- = A_t P_t A_t^\top + Q_t. \quad (5.11) \end{aligned}$$

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1 procedure FILTER( $m_{t-1}, P_{t-1}, A, Q, H, R, y$ )
2    $m_t^- = Am_{t-1}$  // predictive mean
3    $P_t^- = AP_{t-1}A^\top + Q$  // predictive covariance
4    $z = y - Hm_t^-$  // residual
5    $S = HP_t^-H^\top + R$  // innovation covariance
6    $K = P_t^-H^\top S^{-1}$  // gain
7    $m_t = m_t^- + Kz$  // updated mean
8    $P_t = (I - KH)P_t^-$  // updated covariance
9   return ( $m_t, P_t$ ), ( $m_t^-, P_t^-$ )
10 end procedure

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**Algorithm 5.1:** Single step of the Kálmán filter. Variables indexed by  $t$  are outputs, while the variables without indices are only used internally, can thus be overwritten. If the latent state is of dimensionality  $x_t \in \mathbb{R}^L$  and the observations have dimensionality  $y_t \in \mathbb{R}^V$ , then an individual step of the filter has computational complexity  $\mathcal{O}(L^3 + V^3)$  (for the inversion of  $S$  in line 6, and the matrix-matrix multiplications in the latent states in line 3, respectively).

The update, Eq. (5.3) becomes

$$p(x_t \mid y_{0:t}) = \mathcal{N}(x_t; m_t, P_t),$$

where the parameters of this distribution can be computed using the following procedure, giving explicit names to intermediate terms (capital letters denote matrices, lower case vectors):

$$\begin{aligned}
 z_t &:= y_t - H_t m_t^- && \text{(innovation residual),} && (5.12) \\
 S_t &:= H_t P_t^- H_t^\top + R_t && \text{(innovation covariance),} \\
 K_t &:= P_t^- H_t^\top S_t^{-1} && \text{(gain),} \\
 m_t &:= m_t^- + K_t z_t, \\
 P_t &:= (I - K_t H_t) P_t^-. && && (5.13)
 \end{aligned}$$

Many of these terms originate from the field of control theory, where they were made popular by Rudolf Kálmán (1960). The whole prediction update scheme above is thus widely known as the *Kálmán filter*, and the quantity  $K_t$  is called the *Kálmán gain* in his honour.<sup>6</sup>

The “full” posterior marginals – those arising from all observations, in the past and the future – are also Gaussian,

$$p(x_t \mid y) = \mathcal{N}(x_t; m_t^s, P_t^s). \quad (5.14)$$

Their form can be written, from Eq. (5.7), using the parameters

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1 procedure SMOOTHER( $m_t, P_t, A, m_{t+1}^-, P_{t+1}^-, m_{t+1}^s, P_{t+1}^s$ )
2    $G = P_t A^\top (P_{t+1}^-)^{-1}$  // gain
3    $m_t^s = m_t + G(m_{t+1}^s - m_{t+1}^-)$  // posterior mean
4    $P_t^s = P_t + G(P_{t+1}^s - P_{t+1}^-)G^\top$  // posterior covariance
5 end procedure

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**Exercise 5.3** (easy). Using the basic properties of Gaussians from Eqs. (3.4), (3.8) & (3.10) and the prediction-update Eqs. (5.2) & (5.3), show that Eqs. (5.10) to (5.13) hold.

<sup>6</sup> The Kálmán gain is also known as the *optimal* gain for historical reasons (because it is the optimal sensitivity of an electrical filter designed to follow the dynamical system defined above). In our context, this name is redundant. Given the state-space model, the posterior is not just the optimal, but really the only meaningful probabilistic estimator.

**Algorithm 5.2:** Single step of the RTS smoother. Notation as in Algorithm 5.1. The smoother, since it does not actually touch the observations  $y_t$ , has complexity  $\mathcal{O}(L^3)$ .

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1 procedure PREDICT( $m_0, P_0, [A_t, Q_t, H_t, y_t, R_t]_{t=1, \dots, \infty}$ )
2   for  $t = 1, \dots$  do
3      $(m_t, P_t) = \text{FILTER}(m_{t-1}, P_{t-1}, A_t, Q_t, H_t, R_t, y_t)$ 
4   end for
5 end procedure

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of the prediction and updated beliefs, as follows:

$$\begin{aligned}
 &\text{Let } G_t := P_t A_t^\top (P_{t+1}^-)^{-1}, && \text{(smoother gain)} \\
 &\text{then } m_t^s = m_t + G_t (m_{t+1}^s - m_{t+1}^-) \\
 &\text{and } P_t^s = P_t + G_t (P_{t+1}^s - P_{t+1}^-) G_t^\top. && (5.15)
 \end{aligned}$$

These update rules are often simply called the “Kálmán smoother” or, more precisely, the (fixed interval) *Rauch–Tung–Striebel (RTS) smoother* equations.<sup>7</sup> The estimates computed by the Kálmán filter and smoother are depicted in Figure 5.2.

For reference, Algorithms 5.1 and 5.2 summarise the above results in pseudo-code, providing the individual steps for both the filter and the smoother. The wrapper algorithm PREDICT (Algorithm 5.3) solves the task of continuously predicting the subsequent state of a time series. For a finite time series running from  $t = 0$  to  $t = T$ , the algorithm INFER (Algorithm 5.4) returns posterior marginals for every time step  $t$ . Since these marginals (that is, their means and variances) are exactly equal to those of GP regression (§4.2.2), algorithm INFER (Algorithm 5.4) is nothing but a linear-time implementation of GP regression with Markov priors.<sup>8</sup>

The Kálmán filter (and smoother) are so efficient that they can be sometimes applied even if the linearity or Gaussianity of the dynamic or measurement model are violated.<sup>9</sup> In numerics, the case for such fast-and-Gaussian methods is even stronger

**Algorithm 5.3:** Algorithmic wrapper around Algorithm 5.1 to perform on-line prediction at a sequence of points  $t_i$ . PREDICT has constant memory requirements over time, and its computation cost is constant per time step. Thus inference from  $N$  observations requires  $N$  steps of Algorithm 5.1, at cost  $\mathcal{O}(N(L^3 + V^3))$ . See the *Bayesian ODE filter* (Algorithm 38.1); one of its special cases, the EKF0, is analogous to this Algorithm 5.3 (as discussed in §38.3.4).

<sup>7</sup> Rauch, Striebel, and Tung (1965)

<sup>8</sup> Särkkä and Solin (2019), §12.4

<sup>9</sup> Särkkä (2013), §13.1

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1 procedure INFER( $m_0, P_0, [A_t, Q_t, H_t, y_t, R_t]_{t=1, \dots, T}$ )
2   for  $t = 1 : 1 : T$  do
3      $((m_t, P_t), (m_t^-, P_t^-)) =$ 
4        $\text{FILTER}(m_{t-1}, P_{t-1}, A_{t-1}, Q_{t-1}, H_t, R_t, y_t)$ 
5   end for
6   for  $t = T - 1 : -1 : 0$  do
7      $(m_t^s, P_t^s) =$ 
8        $\text{SMOOTHER}(m_t, P_t, A_t, m_{t+1}^-, P_{t+1}^-, m_{t+1}^s, P_{t+1}^s)$ 
9   end for
10 end procedure

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**Algorithm 5.4:** Algorithmic wrapper around Algorithms 5.1 and 5.2 to perform inference for a time-series of finite length. While the runtime of this routine is also linear in  $T$ , in contrast to Algorithm 5.3, it has linearly growing memory cost, to store all the parameters of the posterior and intermediate distributions. See the *Bayesian ODE smoother* (Algorithm 38.2); one of its special cases, the EKS0, is analogous to this Algorithm 5.4 (as discussed in §38.3.4).

because (compared with conventional statistics) saved computational budget can be invested in finer discretisations. Thus, all probabilistic numerical methods in this text use Gaussian methods (such as the Kálmán filter and smoother) for probabilistic regression – with the sole exception of the probabilistic ODE solvers presented in §40.3, where a stronger case for non-parametric posteriors can be made due to highly nonlinear dynamics of ODEs. Accordingly, the standard sequential-Monte-Carlo versions of filters and smoothers, known as *particle filters and smoothers*,<sup>10</sup> will not appear before the ODE chapter VI. There, they will be concisely introduced in §38.4.

<sup>10</sup> Doucet, Freitas, and Gordon (2001)

### ► 5.3 Linear Stochastic Differential Equations

For a moment, let us put aside the observations  $\mathbf{y}$  and only consider the prior defined by the dynamic model from Eq. (5.11). The predictive *means* of this sequence of Gaussian variables follow the discrete linear recurrence relation  $m_{t+1} = A_t m_t$ . When solving numerical tasks, the time instances  $t$  will not usually be immutable discrete locations on a regular grid, but values chosen by the numerical method itself, on a continuous spectrum. We thus require a framework creating continuous *curves*  $x(t)$  for  $t > t_0 \in \mathbb{R}$  that are consistent with such linear recurrence equations. For deterministic quantities, *linear differential equations* are that tool: Consider the linear (time-invariant) dynamical system for  $x(t) \in \mathbb{R}^N$ ,

$$\frac{dx(t)}{dt} = Fx(t); \quad \text{and assume } x(t_0) = x_0. \quad (5.16)$$

This initial value problem is solved by the curve<sup>11</sup>

$$x(t) = \exp(F(t - t_0))x_0.$$

Thus, the linear ordinary differential equation (5.16), together with a set of discrete time locations  $[t_0, t_1, \dots, t_N = T]$  gives rise to the linear recurrence relation

$$x_{t_{i+1}} = A_{t_i} x_{t_i} \quad \text{with} \quad A_{t_i} = \exp(F(t_{i+1} - t_i)).$$

For clarity, the solution to a linear time-invariant ODE at discrete time locations is a linear recurrence relation. This does *not* mean that every recurrence relation can be identified with the solution of one particular linear ODE, or in fact with any ODE at all. But one way to construct recurrence relations is all we need to build interesting probabilistic inference algorithms.

<sup>11</sup> This uses the *matrix exponential*

$$e^X \equiv \exp(X) := \sum_i \frac{X^i}{i!}, \quad (5.17)$$

where  $X^i := \underbrace{X \cdots X}_{i \text{ times}}$  is the  $i$ th power of  $C$  (defining  $C^0 = I$ ). The exponential exists for every complex-valued matrix. Among its properties are:

⊖  $e^0 = I$  (for the zero matrix);

⊖ if  $XY = YX$ , then

$$e^X e^Y = e^Y e^X = e^{X+Y}.$$

Thus, every exponential is invertible:

$$(e^X)^{-1} = e^{-X};$$

⊖ if  $Y$  is invertible, then

$$e^{YXY^{-1}} = Y e^X Y^{-1},$$

$$\text{thus, } \det e^X = e^{\text{tr} X};$$

⊖ if  $D = \text{diag}_i(d_i)$  then  $e^D = \text{diag}_i(e^{d_i})$  (using the scalar exponential). In particular, if  $X = VDV^{-1}$  is the eigen-decomposition of  $X$ , then

$$e^X = V e^D V^{-1}$$

provides a practical way to compute the exponential of diagonalisable matrices. Another important way to compute matrix exponentials is if  $X$  is *nilpotent* (i.e. there exists an  $m \in \mathbb{N}$  such that  $X^m = 0$ ), in which case the exponential can be explicitly computed by Eq. (5.17).



The construction of probabilistic models for continuous time series requires a generalisation of the above construction for deterministic linear dynamic systems to stochastic processes: We need a way to build the Gaussian density around the mean prediction  $m_{t_{i+1}} = A_{t_i} m_{t_i}$ , such that the conditional distribution has the form of Eq. (5.10). In particular, given  $x_{t_i} = m_{t_i}$ , we would like a relative, or “differential” way to add a centred Gaussian disturbance  $\xi_t \sim \mathcal{N}(0, Q_t)$  over the time interval  $[t_i, t_{i+1}]$ .

Such a generalisation is provided by *linear stochastic differential equations* (linear SDEs). Inconveniently, the stochastic paths that solve such “differential” equations are a.s. nowhere differentiable and thus have a.s. infinite total variation.<sup>12</sup> A rigorous definition of SDEs thus requires a new integral w.r.t. such a stochastic process (the *Itô integral*) which gives rise to a new kind of calculus. To avoid its complicated introduction, we will, however, only provide an intuitive definition of SDEs. The full-blown theory can, e.g., be found in the detailed book by Karatzas and Shreve (1991) or in the more concise one by Oksendal (2003); if slightly less rigour is acceptable to the reader, we recommend the more accessible book by Särkkä and Solin (2019).

<sup>12</sup> Oksendal (2003), §3.1

Above, general Gaussian processes were defined as an infinite-dimensional generalisation of finite-dimensional distributions. In similar fashion, we will define a linear SDE as the abstract object giving rise to Gaussian probabilistic recurrence relations for finite time steps.

**Definition 5.4** (Linear SDE).<sup>13</sup> Consider curves  $x : t \in \mathbb{R} \mapsto x(t) \in \mathbb{R}^N$  for  $t > t_0$ . Assume matrices  $F \in \mathbb{R}^{N \times N}$  and a vector  $L \in \mathbb{R}^N$ . For the purposes of this text, the linear time-invariant stochastic differential equation

$$dx(t) = Fx(t) dt + L d\omega_t$$

(see below for an explanation of the notation  $d\omega_t$ ), together with the initial value  $x(t_0) = x_0$ , describes the local behaviour of a unique Gaussian process, determined by the following mean and covariance function:

$$\mathbb{E}(x(t)) = e^{F(t-t_0)} x_0, \quad (5.18)$$

$$\begin{aligned} \text{cov}(x(t), x(t')) &= \int_{t_0}^{\min t, t'} e^{F(t-\tau)} L L^\top e^{F^\top(t'-\tau)} d\tau \\ &=: k(t, t'). \end{aligned} \quad (5.19)$$

<sup>13</sup> In addition to sidestepping complications involved with general stochastic processes on the real line, the simplified definition for SDEs chosen here also attempts to avoid confusion with the definition of probabilistic algorithms for the solution of deterministic ordinary differential equations in Chapter VI. The aim in that chapter will be to cast the solution of ODEs as the construction of linear SDEs that concentrate as much probability mass as possible around the true solution. It will *not* be the aim to construct more involved stochastic processes driven by the ODE (and its unknown solution) itself.

This process is known as the solution of the SDE. In particular, this

gives rise to the discrete time stochastic recurrence relation

$$p(x_{t_{i+1}} | x_{t_i}) = \mathcal{N}(x_{t_{i+1}}; A_{t_i} x_{t_i}, Q_{t_i}), \quad (5.20)$$

with

$$\begin{aligned} A_{t_i} &:= \exp(F(t_{i+1} - t_i)) \quad \text{and} \\ Q_{t_i} &:= \int_0^{t_{i+1} - t_i} e^{F\tau} L L^T e^{F^T \tau} d\tau. \end{aligned} \quad (5.21)$$

The most basic choice  $N = 1, L = 1, F = 0$  yields the SDE “ $dx(t) = d\omega_t$ ” and thus an implicit definition of that ominous object  $d\omega_t$ , which is called the *increment of the Wiener process*. The solution of this SDE has (since  $e^0 = 1$ ), the constant mean function  $\mathbb{E}(x(t)) = \mu(t) = x_0$  (i.e. if we set  $x_0 = 0$ , vanishing mean), and the covariance function is  $k(t, t') = \min(t, t') - t_0$ . This solution is thus the *Wiener process*<sup>14</sup> with starting time  $t_0$  (see Eq. (4.5)), which is depicted in Figure 5.3 for  $t_0 = 0$ . The mysterious  $d\omega_t$  is, strictly speaking, nothing but a placeholder for an Itô integral; i.e. the SDE “ $dx = d\omega_t$ ” is in essence just a shorthand for the Itô stochastic integral equation  $x(t) = \int_0^t d\omega_s$ .<sup>15</sup> Thus  $d\omega_t/dt$  is, loosely speaking, some kind of “derivative of the Wiener process”; it is hence often thought of as Gaussian *white noise*. While this is a useful intuition, it is completely informal because a sample path from the Wiener process  $\omega_t$  is almost surely not differentiable, almost everywhere. Figure 5.4 gives an intuition for state-space inference using the Wiener process itself as the prior measure.

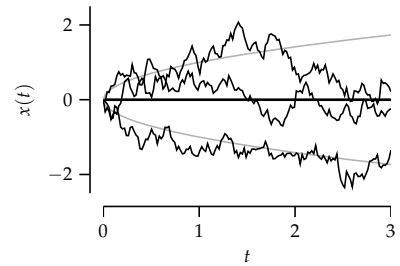
## ► 5.4 Polynomial Splines

An interesting class of linear SDEs arises from *integrals* of the Wiener process. This can be achieved by considering the SDE with  $F \in \mathbb{R}^{(q+1) \times (q+1)}$  and  $L \in \mathbb{R}^{(q+1)}$  as

$$F = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & \cdots & 0 & 0 \end{bmatrix}, \text{ and } L = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ \theta \end{bmatrix}. \quad (5.22)$$

This structure<sup>16</sup> ensures that the elements of  $x$  are derivatives of each other. So they can be interpreted as derivatives of a function  $f: \mathbb{R} \rightarrow \mathbb{R}$ :

$$x(t) = \begin{bmatrix} f(t) & f'(t) & f''(t) & \cdots & f^{(q)}(t) \end{bmatrix}. \quad (5.23)$$



**Figure 5.3:** Draws from the Wiener process  $\omega_t$  (here with  $x_{t=0} = 0$ ) are paths of Brownian motion: continuous, but non-differentiable, almost everywhere. Their expected deviation from  $x_0$  (grey lines) grows as  $\sqrt{t}$ .

<sup>14</sup> Candidates for the invention of the Wiener process include the Danish astronomer Thiele in 1880 (see a review by Lauritzen in 1981), the French mathematician Bachelier, the Austrian physicist Smoluchowski, and Albert Einstein with his *Annus Mirabilis* paper on Brownian motion (Einstein, 1906). Wiener (1923), however, was arguably the first to show the existence of the process. An impressive encyclopedic handbook on the Wiener process was provided by Borodin and Salminen (2002).

<sup>15</sup> See §5.2.A. in Karatzas and Shreve (1991) for more information on the stochastic-integral form of SDEs.

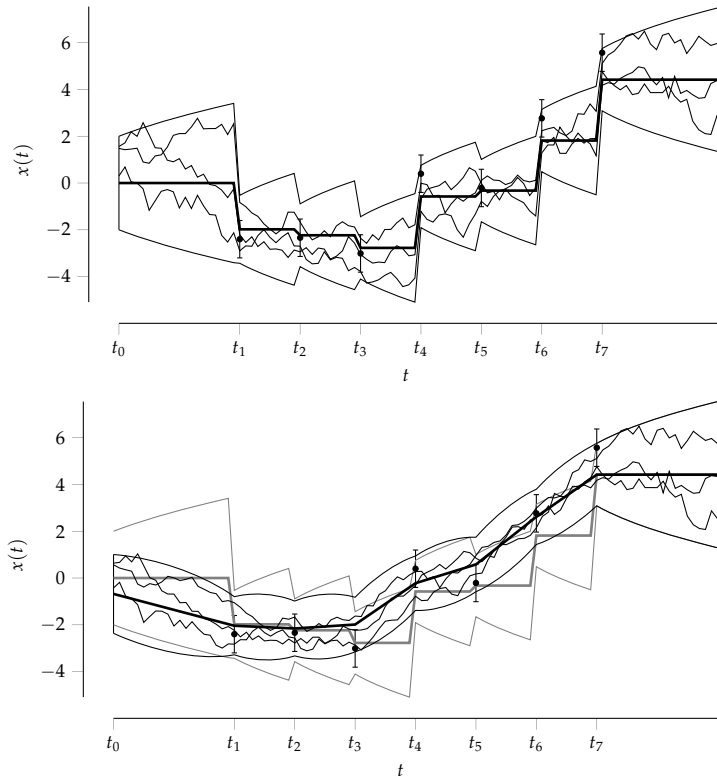
<sup>16</sup> Another common notation if the step size is fixed ( $t_{i+1} - t_i = h \forall i$ ) is to scale the state space to  $\tilde{x}(t) = Bx(t)$  with  $B = \text{diag}(1, h, \dots, h^q/q!)$ , and a corresponding  $\tilde{F} = BFB^{-1}$ ,  $\tilde{L} = BL$  (this means  $F_{i,i+1} = i/h$ ). In this notation, Eq. (5.23) becomes

$$\tilde{x}_i(t) = \frac{h^{i-1} f^{(i-1)}(t)}{(i-1)!}.$$

The advantage is that the resulting  $\tilde{A}$  and  $\tilde{Q}$  have much simpler form:  $\tilde{A}$  becomes the *Pascal triangle* matrix (the upper triangular matrix containing the Pascal triangle), and the elements of  $Q$  depend on  $h$  only in a scalar fashion:

$$\begin{aligned} [\tilde{A}]_{ij} &= \mathbb{I}(j \geq i) \binom{j}{i}, \\ [\tilde{Q}]_{ij} &= \frac{\theta^2 h^{2q+1}}{(2q+3-i-j)(q+1-i)!} \\ &\quad \times \frac{1}{(q+1-j)!(i-1)!(j-1)!}. \end{aligned}$$

This can help both with numerical stability and efficient implementations.



**Figure 5.4:** State-space inference using the most elementary Wiener process prior, i.e. the linear time invariant model with  $F = 0, L = 1, H = 1$ . Observations shown as black circles with error bars at two standard deviations of the Gaussian observation noise. Top: Filtering. Predictive means in solid black, two standard deviations in thin black. Three joint samples from the predictive distribution, constructed in linear time during the prediction run, are plotted in thin black, also. Bottom: Posterior distribution after smoothing (filtering distribution in grey for comparison). The posterior samples (which are indeed valid draws from the joint posterior) are produced by taking the samples from the predictive distribution and scaling/shifting them (deterministically) during the smoothing run; they do not involve additional random numbers.

The corresponding discrete-time objects  $A$  and  $Q$  contain elements that are polynomials of the time steps  $h_i := t_{i+1} - t_i$ . The matrix  $A(h)$  is upper triangular (and  $Q$ , of course, is SPD), with elements<sup>17</sup> ( $1 \leq i, j \leq q + 1$ )

$$[A(h)]_{ij} = [\exp(Fh)]_{ij} = \mathbb{I}(j \geq i) \frac{h^{j-i}}{(j-i)!}, \quad (5.25)$$

$$[Q(h)]_{ij} = \theta^2 \frac{h^{2q+3-i-j}}{(2q+3-i-j)(q+1-i)!(q+1-j)!}. \quad (5.26)$$

A detailed derivation of  $A$  and  $Q$  is provided in the literature.<sup>18</sup> The corresponding covariance functions have a tedious form, but they are also polynomials. For example, for  $q = 1$  – the *integrated Wiener process* – the kernel is (for  $t_0 = 0$ )

$$\text{cov}(f(a), f(b)) = \theta^2 \left( \frac{1}{3} \min^3(a, b) + |a - b| \frac{1}{2} \min(a, b) \right) \quad (5.27)$$

(see also Figure 4.3). We recall that the posterior mean function of a Gaussian process is a weighted sum of kernel evaluations. Hence, under the choice (5.27) and noise-free observations of  $f$ , the posterior mean on  $f$  would be the piecewise cubic interpolant of the data (which is unique within the convex hull of the

<sup>17</sup> Since  $\exp(X)^{-1} = \exp(-X)$ , Eq. (5.25) also immediately provides

$$[A^{-1}(h)]_{ij} = \mathbb{I}(j \geq i) \frac{(-h)^{j-i}}{(j-i)!}. \quad (5.24)$$

<sup>18</sup> Kersting, Sullivan, and Hennig (2020), Appendix A; note that zero-based indexing is used there.

data). It is the *cubic spline*. More generally, posterior interpolants of the  $q$ -times integrated Wiener process are  $2q + 1$ -ic splines.

## ► 5.5 The Matérn Family

Another important base case is given by the univariate SDE (i.e.  $x(t) \in \mathbb{R}$ ) with a negative  $F$ . For convenience further below, we will use the parametrisation  $F = -1/\lambda$  and  $L = 2\theta/\sqrt{\lambda}$  for  $\theta, \lambda \in \mathbb{R}_+$ . This setup captures the velocity of a gas particle of temperature  $\theta/\sqrt{\lambda}$  in a linear (harmonic) potential well of force constant  $1/\lambda$ , and its solution is known as the *Ornstein–Uhlenbeck* (OU) process.<sup>19</sup> The SDE

$$dx = -\frac{x}{\lambda} dt + \frac{2\theta}{\sqrt{\lambda}} d\omega_t$$

with  $x(t_0) = x_0$  yields, with Eqs. (5.18) and (5.19),

$$\mathbb{E}(x(t)) = x_0 e^{-\frac{t-t_0}{\lambda}} \quad \text{and} \quad k(x, x') = \theta^2 \left( e^{-\frac{|t-t'|}{\lambda}} - e^{-\frac{t+t'-2t_0}{\lambda}} \right).$$

The special case of  $t_0 \rightarrow -\infty$  is the stationary limit of the process. It has vanishing mean function, and covariance

$$k(t, t') = \theta^2 e^{-\frac{|t-t'|}{\lambda}}.$$

General integrals of the OU process form an important class of regression models in statistics and machine learning. The *Matérn*<sup>20</sup> class of covariances<sup>21</sup> is given, using the distance  $r := |t - t'|$  by

$$k_\nu(r) = \theta^2 \frac{2^{1-\nu}}{\Gamma(\nu)} \left( \sqrt{2\nu} \frac{r}{\lambda} \right)^\nu K_\nu \left( \sqrt{2\nu} \frac{r}{\lambda} \right),$$

where  $K_\nu$  is the modified Bessel function. For  $\nu = q + 1/2$  with integer  $q$ , these kernels have the slightly more concrete form<sup>22</sup>

$$k_{q+1/2}(r) = \theta^2 \frac{\Gamma(q+1)}{\Gamma(2q+1)} \sum_{i=0}^q \frac{(q+i)!}{i!(q-i)!} \left( \sqrt{8\nu} \frac{r}{\lambda} \right)^{q-i} \cdot \exp \left( -\sqrt{2\nu} \frac{r}{\lambda} \right).$$

Or yet more explicitly, for the first three integer (and most popular) choices  $q = 0, 1, 2$ :

$$\begin{aligned} k_{1/2}(r) &= \theta^2 \cdot \exp \left( -\frac{r}{\lambda} \right), \\ k_{3/2}(r) &= \theta^2 \left( 1 + \frac{\sqrt{3}r}{\lambda} \right) \cdot \exp \left( -\frac{\sqrt{3}r}{\lambda} \right), \\ k_{5/2}(r) &= \theta^2 \left( 1 + \frac{\sqrt{5}r}{\lambda} + \frac{5r^2}{3\lambda^2} \right) \cdot \exp \left( -\frac{\sqrt{5}r}{\lambda} \right). \end{aligned}$$

<sup>19</sup> Uhlenbeck and Ornstein (1930)

<sup>20</sup> Matérn (1960)

<sup>21</sup> The naming of the family is due to Stein (1999). See also Rasmussen and Williams (2006), Eq. (4.16).

<sup>22</sup> Readers with a background in functional analysis might find it helpful to know that the RKHS arising from the Matérn kernel  $k_{q+1/2}$  is norm-equivalent to the Sobolev space  $H^{q+1}(\mathbb{R})$ . See, e.g., Kanagawa, Sriperumbudur, and Fukumizu (2020).

The associated Gaussian processes are solutions of multivariate state-space models of the form<sup>23</sup>

$$dz(t) = Fz(t) dt + L d\omega_t$$

with  $z \in \mathbb{R}^{q+1}$  and

$$F = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & 0 & \vdots \\ \vdots & & \ddots & \ddots & 0 \\ 0 & & & 0 & 1 \\ -a_0 & -a_1 & \cdots & -a_{q-1} & -a_q \end{bmatrix} \quad \text{and} \quad L = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ \theta \end{bmatrix}, \quad (5.28)$$

where the  $a_i$  are the coefficients of the polynomial  $(\xi + i\omega)^{-(q+1)}$  with  $\xi := \sqrt{2\nu}/\lambda$ .<sup>24</sup> For example, for the three values  $q = 0, 1, 2$  from above, we get

$$dz(t) = -\xi z(t) dt + \theta d\omega_t, \quad (5.29)$$

$$dz(t) = \begin{bmatrix} 0 & 1 \\ -\xi^2 & -2\xi \end{bmatrix} z(t) dt + \begin{bmatrix} 0 \\ \theta \end{bmatrix} d\omega_t$$

$$dz(t) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -\xi^3 & -3\xi^2 & -3\xi \end{bmatrix} z(t) dt + \begin{bmatrix} 0 \\ 0 \\ \theta \end{bmatrix} d\omega_t. \quad (5.30)$$

A more general and detailed introduction to this family of stochastic processes is provided by Särkkä and Solin (2019).

## ► 5.6 Riccati Equations and Steady State

In numerical computations, we will often run a Gaussian filter for many steps through time. The question will then arise whether the algorithm will actually be able to identify an object of interest to finite uncertainty, or whether it might “lose track” over time, in the sense that the posterior predictive covariance  $P_i^-$  rises without bound. If this is not the case, does  $P_i^-$  converge to some finite value (and which one) for  $i \rightarrow \infty$ ?

There is an elegant framework for this purpose, which is well-studied in control engineering,<sup>25</sup> where Kálmán filters are a foundational tool and the ability to track and control a system is of central concern. Consider a linear time-invariant system defined by the parameters  $A, Q, H, R$ . From the prediction (Eq. (5.11)) and update (Eq. (5.13)), the step from  $P_i^-$  to  $P_{i+1}^-$  is given by the *deterministic* step (since it does not involve the observation  $y$ )

<sup>23</sup> Hartikainen and Särkkä (2010); Särkkä, Solin, and Hartikainen (2013).

<sup>24</sup> This matrix  $F$  is known as the *companion matrix* of the polynomial

$$p(x) = a_0 + a_1 x^1 + \cdots + a_q x^q + x^{q+1},$$

because  $p$  is both the characteristic and the minimal polynomial of  $F$ . More in §7.4.6 of Golub and Van Loan (1996), and on pp. 405ff. of Wilkinson (1965).

**Exercise 5.5** (moderate, solution on p. 360). Find explicit forms for the matrices  $A_t, Q_t$  (Eq. (5.21)) associated with the discrete-time forms of Eqs. (5.29)–(5.30).

<sup>25</sup> Bougerol (1993)

**Exercise 5.6** (moderate, solution on p. 360). Equation (5.31) demonstrates that the predictive covariances  $P_i^-$  are governed by a discrete time algebraic Riccati equation. Show that the evolution of the estimation covariances  $P_i$  and the smoothed covariances  $P_i^s$  are also described by DAREs.

$$P_{i+1}^- = AP_i^- A^\top - AP_i^- H^\top (HP_i^- H^\top + R)^{-1} HP_i^- A^\top + Q \quad (5.31)$$

$$= Q + A((P_i^-)^{-1} + H^\top R^{-1} H)^{-1} A^\top.$$

Equations of this form are known as *discrete-time algebraic Riccati equations* (DAREs).<sup>26</sup> Whether a DARE has a fix point, and if so, where that fix point lies, can be answered by considering the following *symplectic matrix*:<sup>27</sup>

$$Z = \begin{bmatrix} A^\top + H^\top R^{-1} H A^{-1} Q & -H^\top R^{-1} H A^{-1} \\ -A^{-1} Q & A^{-1} \end{bmatrix}.$$

If  $Z$  has no eigenvalues on the unit circle, then exactly half its eigenvalues are inside the unit circle, and the iteration (5.31) converges to a finite fix-point, known as the *steady-state* predictive covariance. To find it, consider the matrix  $U \in \mathbb{C}^{2N \times 2N}$  of the eigenvectors of  $Z$ . Assume that they are sorted such that the first  $N$  columns of  $U$  correspond to the eigenvectors for eigenvalues *inside* the unit circle (and the other  $N$  columns to those *outside* the circle). Then separate  $U$  into square  $N \times N$  sub-matrices as

$$U = \begin{bmatrix} U_1 & U_3 \\ U_2 & U_4 \end{bmatrix}.$$

The steady-state predictive covariance is given by

$$P_\infty^- = U_2 U_1^{-1}.$$

This solution is relatively easy to state, but it is not the numerically most efficient way to find  $P_\infty^-$ . Advanced algorithms can be found in the book by Faßbender (2007). More generally, Riccati equations have been the subject of deep analysis. A classic book was written by Reid (1972). Bini, Iannazzo, and Meini (2011) provide a more recent review.

<sup>26</sup> The name reflects that, under some regularity assumptions, there is a continuous time limit given by the matrix differential equation

$$dP = AP + PA^\top - PH^\top R^{-1} HP + Q \quad (5.32)$$

(in the sense that the sequence of  $P$ 's produced by Eq. (5.31) are solutions of Eq. (5.32)). Historically, quadratic ordinary differential equations

$$x'(t) = q(t) + a(t)x(t) + b(t)x^2(t)$$

(of which Eq. (5.32) is a matrix-valued generalisation) are named after the eighteenth-century work of Jacopo Riccati (1724).

<sup>27</sup> A symplectic matrix  $M$  is a matrix that satisfies  $M^\top \Omega M = \Omega$  for some fixed non-singular, skew-symmetric matrix  $\Omega$ . In our case, e.g.,

$$\Omega = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}.$$

Symplectic matrices have unit determinant. They are invertible, with  $M^{-1} = \Omega^{-1} M^\top \Omega$ , and a product of two symplectic matrices is a symplectic matrix. Hence they form a group. A Lie group, in fact, known as the symplectic group. Its generators are the *Hamiltonian matrices*, matrices of the form

$$X = \begin{bmatrix} A & B \\ C & -A^\top \end{bmatrix}$$

with symmetric matrices  $B, C$ . There is a corresponding stability analysis for the continuous-time algebraic Eq. (5.32) involving a Hamiltonian matrix.